



UNIVERSITY OF MINHO
SCHOOL OF ENGINEERING

FMdeploy - Simplifying Machine Learning Deployment for Academia via Web Interfaces

Dissertation
Integrated Master's in Biomedical Engineering

Supervisor
Victor Alves

Marcelo Marreiros
December 4th, 2023





Table of contents

01

Introduction

Context, Motivation &
Objectives

02

FMdeploy

Architecture & Features

03

ML Models

Deployed ML Models

04

Conclusions

Findings & Research
Opportunities





01

Introduction

Context, Motivation, and Objectives





Context



Medical Informatics

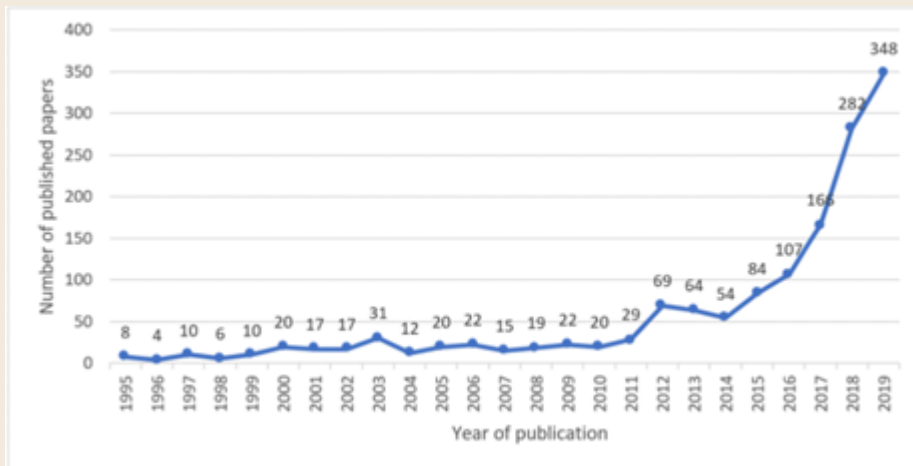
Emerging as a pivotal scientific discipline that handles how best to use information to improve healthcare.



AI in Medicine

Experiencing exponential growth, revolutionizing medical practices through advanced learning methodologies.

Motivation

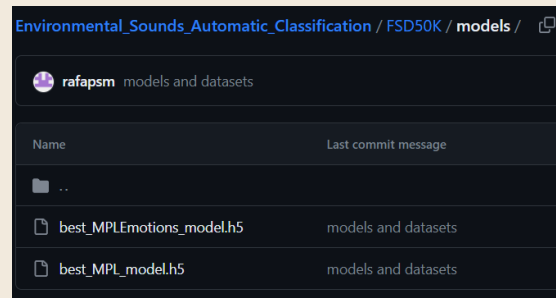
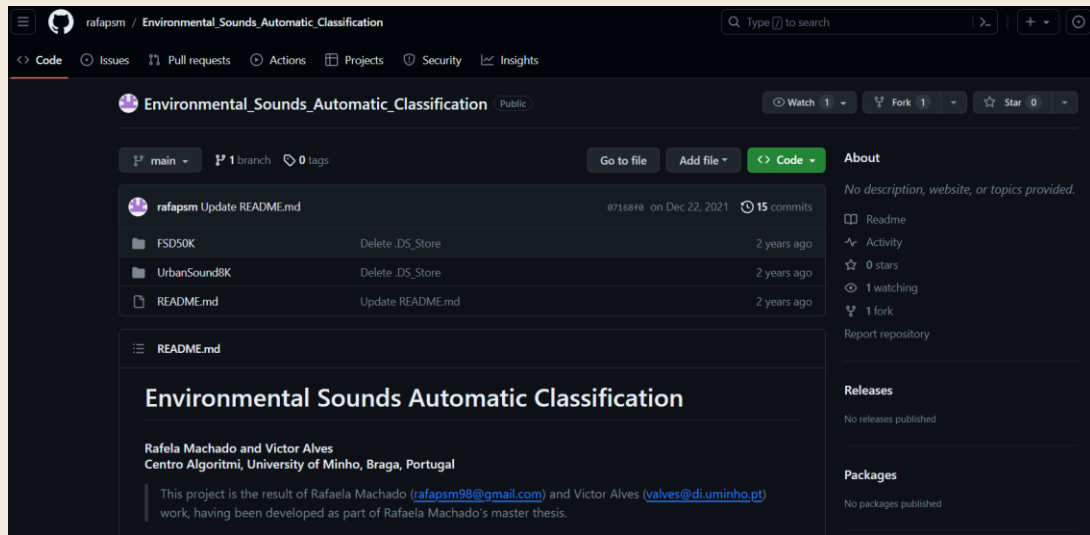


Distribution of bibliographic records per year

Advancements in AI-focused Healthcare Research

Challenges in Translation to Real-World Applications

Motivation



**Importance of Model
Deployment**

**Complexities and Need
for Collaboration**

**Enhancing Acessibility
Through User-Friendly
Interfaces**

Objectives



1

Develop a user-friendly platform to simplify machine learning model deployment.



2

Analyze the ML workflow to define integration processes for ML models.



3

Test the platform by deploying a machine learning model.



4

Conduct iterative development and host a prototype on a server for continuous platform improvement.



02

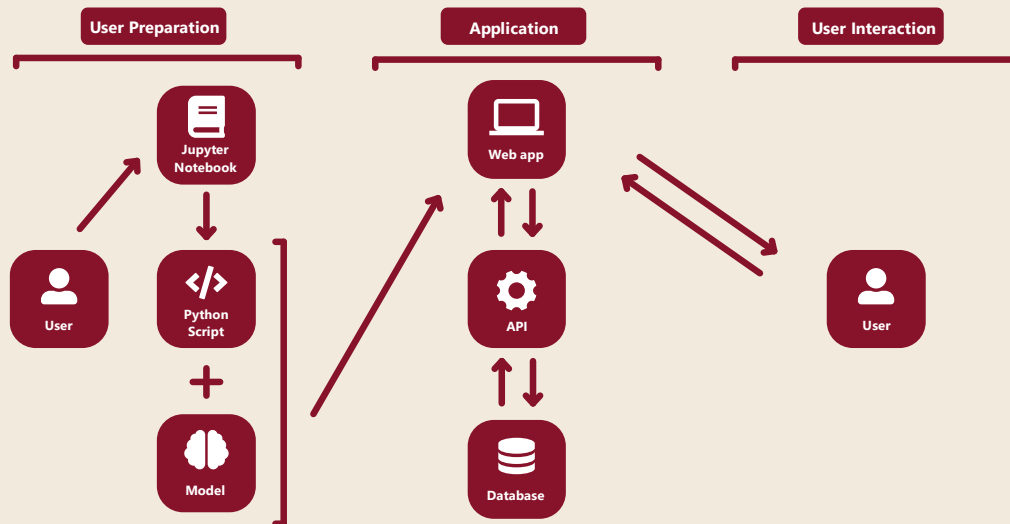
FMdeploy

Architecture and Features

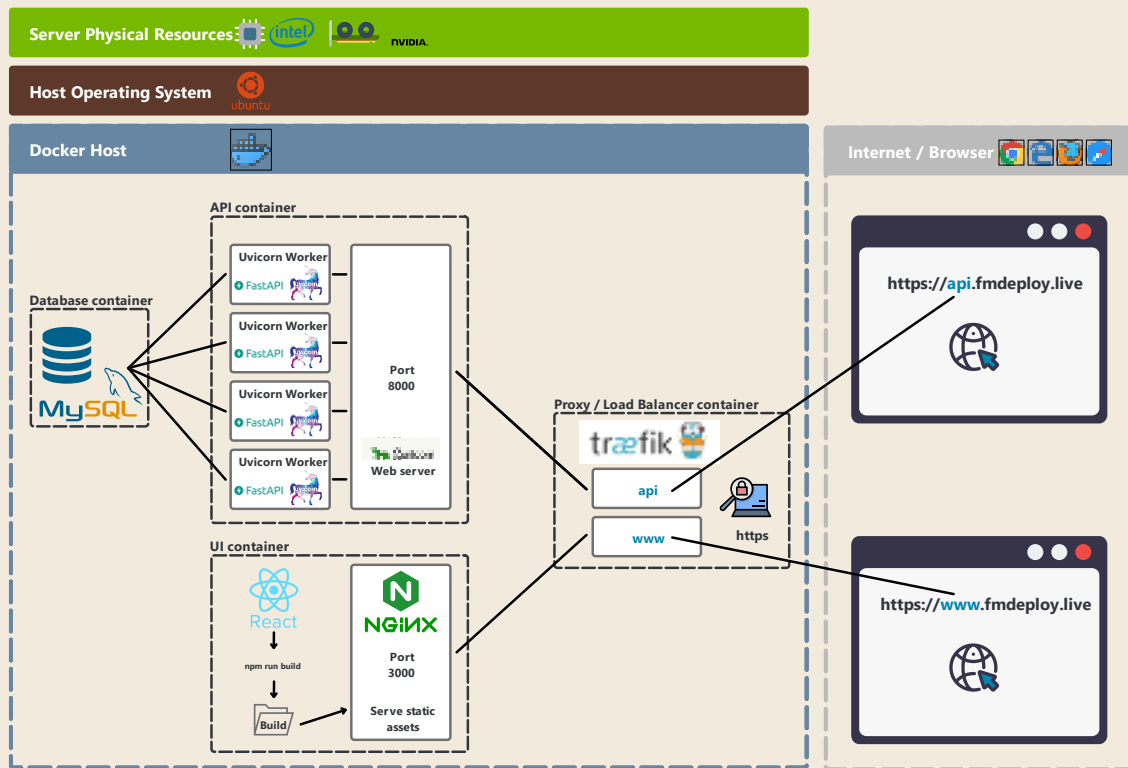


Workflow

```
def load_models(modelpaths):  
    try:  
        (...)  
        return True  
    except:  
        return logging.exception("load_models: ")  
  
def run(input_file_path, output_file_name,  
        output_directory_path):  
    try:  
        (...)  
        return True  
    except:  
        return logging.exception("run: ")
```



Architecture



FMdeploy - FastAPI

Welcome to FMdeploy API !

Developed with [FastAPI](#). To check the documentation please use one of the links bellow.

Docs Swagger UI

Docs ReDoc



FMdeploy

FastAPI 0.10.0 **GA**

[OpenAPI JSON](#)

Authorize

Authentication

POST /login Login User

User

GET /user/is_admin Check if User is Admin

GET /user/admin Get All Users

POST /user/admin Create Administrator

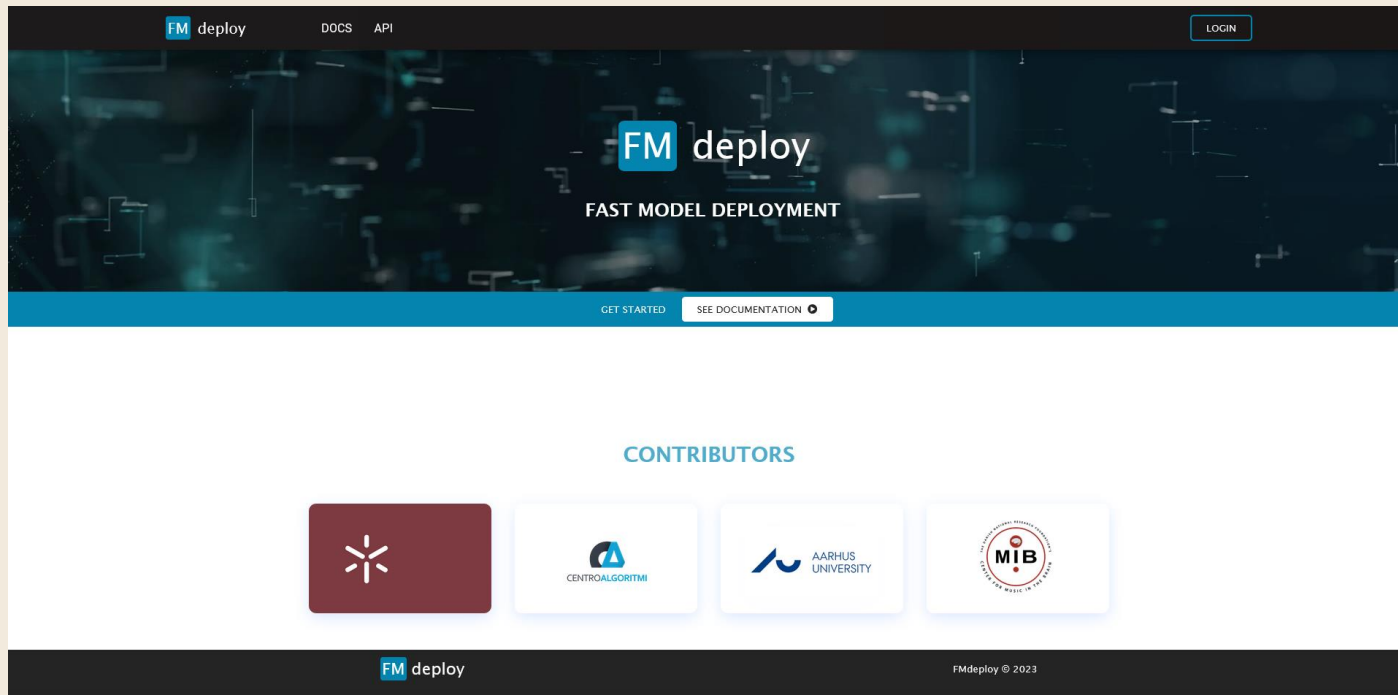
GET /user Get Current User

PUT /user Update Current User

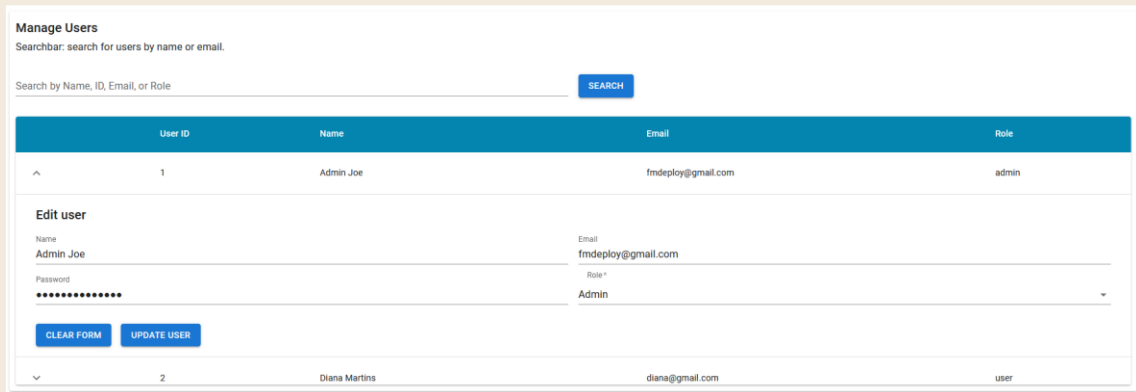
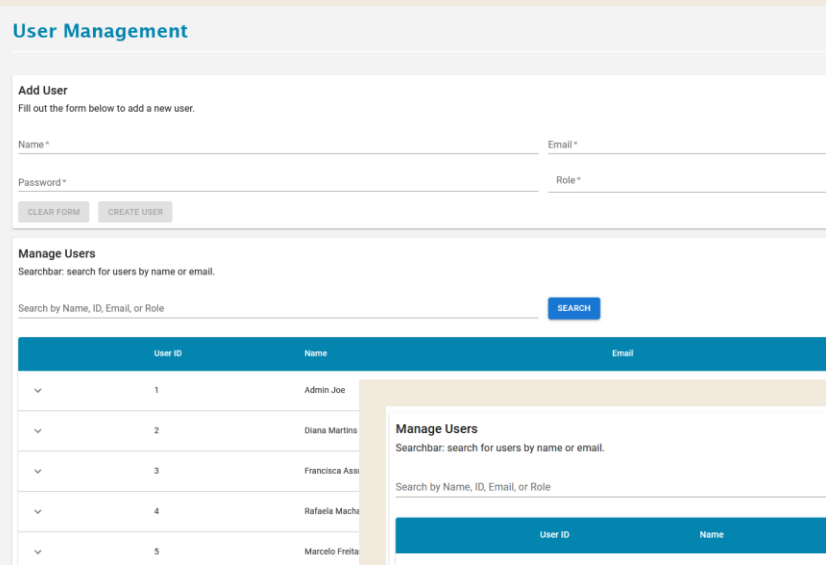
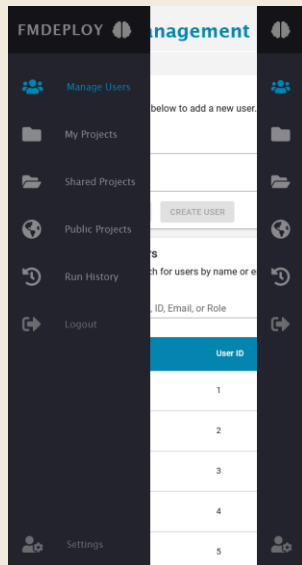
POST /user Create User

DELETE /user Delete Current User

FMdeploy - ReactJS



FMdeploy - ReactJS



A 6x4 grid of dots, consisting of 6 rows and 4 columns of small black dots.



The screenshot displays the Sound Events Explorer interface, which is divided into three main sections: My Projects, Public Projects, and Shared Projects. Each section features a search bar and a list of projects with details such as ID, DATE, and TITLE. The My Projects section also includes a DESCRIPTION field. The interface is designed with a dark sidebar on the left containing navigation icons and a main content area with a light background. The Shared Projects section is highlighted with a yellow background.

My Projects

Search by	ID	DATE	TITLE	DESCRIPTION
	7bb7384e367456b86ad59c005b0e155	10/07/2023	Decoding Sound Events Related to ASD - png	AUTHOR: Diana Martins PRIVATE: false INPUT TYPE: wav OUTPUT TYPE: png DESCRIPTION: A model that classifies environmental sounds related to auditory hypersensitivity in individuals with ASD, using the XGBoost library. Output in png.
	ccb361e6f5c54fc3907fba944c59d23	10/07/2023	Decoding Sound Events Related to ASD - csv	

Public Projects

Search by	ID	DATE	TITLE
	7bb7384e367456b86ad59c005b0e155	10/07/2023	Decoding Sound Events Related to ASD - png
	28df24bb5d0048e89fffc795aa2dbe8	10/07/2023	MP3 to Wav Converter

Shared Projects

Search by	ID	DATE	TITLE
	1322e00a8ed4d168069c437231d7587	10/07/2023	Environmental Sounds Automatic Classification
	425df406fa874dadbcd7bb016588b1bf	10/07/2023	Automatic Segmentation of the Olfactory Bulb
	ccb361e6f5c54fc3907fba944c59d23	10/07/2023	Decoding Sound Events Related to ASD - csv

FMdeploy - ReactJS

New Project

BACK

1

Create Project

2

Add Project Files

3

Result

Title

Description

Input Type

Output Type

☒ Private

NEXT

New Project

BACK

1

Create Project

2

Add Project Files

3

Result

2. Add Project Files

Python Script

ob_masks_fmdeploy.v2.py

CLEAR

Models

Drop Files Here
- or -
Click to Upload

- modelo_CNN_R_treinado_v4_174x190.h5 - 24812272 bytes
- modelo_CNN_treinado_v4.h5 - 48409608 bytes
- modelo_FCN_segmentation_32x32.h5 - 101371640 bytes

CLEAR

BACK SUBMIT

New Project

BACK

1

Create Project

2

Add Project Files

3

Result

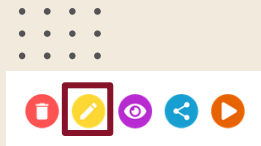
1. Create Project
Project created successfully

2. Submit Python Script
Python Script uploaded successfully

3. Submit Model Files
Model Files uploaded successfully

NEW PROJECT

FMdeploy - ReactJS





EDIT: Decoding Sound Events Related to ASD - png

BACK

PROJECT ID: 7bb7384e6367456b86ad59c005b0e155

AUTHOR: Diana Martins

DATE: 10/07/2023, 15:20:21

DESCRIPTION:

A model that classifies environmental sounds related to auditory hypersensitivity in individuals with ASD, using the XGBoost library. Output in png.

EDIT DATA:

Title

Decoding Sound Events Related to ASD - png

Description

A model that classifies environmental sounds related to auditory hypersensitivity in individuals with ASD, using the XGBoost library. Output in png.

Input Type

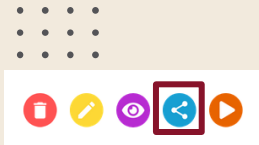
.wav

Output Type

.png

☐ Private

SUBMIT



FMdeploy - ReactJS





SHARE: Decoding Sound Events Related to ASD - csv

BACK

AI ID: cbb361e6f5c54fc3907feba944c59d23

AUTHOR: Diana Martins

INPUT FILE TYPE: .wav

DATE: 10/07/2023, 15:18:02

PRIVATE: true

SHARED WITH:

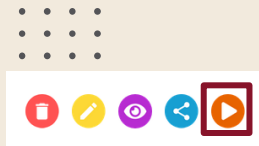
Name	Email	
Marcelo Freitas	marcelo@gmail.com	
Admin Joe	fmdeploy@gmail.com	

SHARE:

Email

SHARE

FMdeploy - ReactJS





RUN: Decoding Sound Events Related to ASD - png

PROJECT ID: 7bb7384e6367456b86ad59c005b0e155

AUTHOR: Diana Martins

INPUT FILE TYPE: .wav

DATE: 10/07/2023

PRIVATE: false

DESCRIPTION:

A model that classifies environmental sounds related to auditory hypersensitivity in individuals with ASD, using the XGBoost library. Output in png.

1. INPUT

Drop File Here
- or -
Click to Upload
(you can only drop 1 file here)

2. OUTPUT

CLEAR

SUBMIT

FMdeploy - ReactJS

1. INPUT

Drop File Here
- or -
Click to Upload
(you can only drop 1 file here)

CLEAR

SUBMIT

2. OUTPUT

"result_forest.png"

DOWNLOAD

FLAG

3. OUTPUT PREVIEW

Y_test	first_pred	first_percent	second_pred	second_percent	third_pred	third_percent
3	2	0.982	0	0.009	1	0.008
3	2	0.993	1	0.003	0	0.002
3	2	0.989	0	0.005	1	0.004
3	2	0.992	1	0.004	0	0.002
3	2	0.99	1	0.004	0	0.004
3	2	0.883	1	0.096	0	0.019
3	2	0.991	0	0.005	1	0.003
3	2	0.991	0	0.004	1	0.003
3	2	0.994	0	0.003	1	0.002
3	2	0.993	1	0.004	0	0.002
3	2	0.993	0	0.003	1	0.002
3	2	0.992	0	0.003	1	0.003
3	2	0.981	1	0.01	0	0.007
3	2	0.992	0	0.004	1	0.003
3	2	0.992	0	0.004	1	0.003

FMdeploy - ReactJS



AUTHOR: Diana Martins

DATE: 10/07/2023

INPUT FILE TYPE: .wav

PRIVATE: false

DESCRIPTION:

A model that classifies environmental sounds related to auditory hypersensitivity in individuals with ASD, using the XGBoost library. Output in png.

1. INPUT

Flag erroneous output

Flag an output if you notice something wrong or unexpected, and describe the issue or error you have observed in the provided text box.

Description

CLEAR

SUBMIT

FLAG

2. OUTPUT

forest.png"

3. OUTPUT PREVIEW

y_test	first_pred	first_percent	second_pred	second_percent	third_pred	third_percent
3	2	0.982	0	0.009	1	0.008
3	2	0.993	1	0.003	0	0.002
3	2	0.989	0	0.005	1	0.004
3	2	0.992	1	0.004	0	0.002
3	2	0.99	1	0.004	0	0.004

FMdeploy - ReactJS



A model that classifies environmental sounds related to auditory hypersensitivity in individuals with ASD, using the XGBoost library.

1. INPUT

Drop File Here
- or -
Click to Upload
(you can only drop 1 file here)

CLEAR

SUBMIT

2. OUTPUT

FLAGGED OUTPUTS

Flag	Project	Name	Email	Input	Output	Time
	Decoding Sound Events Related to ASD - csv	Marcelo Freitas	marcelo@gmail.com	forest.wav	result_forest.csv	11/07/2023, 13:18
	Decoding Sound Events Related to ASD - csv	Marcelo Freitas	marcelo@gmail.com	forest.wav	result_forest.csv	11/07/2023, 17:36

Output Flagged

Flag description: The output file csv is missing one entry.

FMdeploy - ReactJS

Run History

Flag	Project	Name	Email	Input	Output	Time
🚩	Decoding Sound Events Related to ASD - csv	Marcelo Freitas	marcelo@gmail.com	forest.wav	result_forest.csv	11/07/2023, 17:36
Output Flagged Flag description: The output file csv is missing one entry.						
✓	Decoding Sound Events Related to ASD - csv	Marcelo Freitas	marcelo@gmail.com	forest.wav	result_forest.csv	11/07/2023, 13:25
🚩	Decoding Sound Events Related to ASD - csv	Marcelo Freitas	marcelo@gmail.com	forest.wav	result_forest.csv	11/07/2023, 13:18
🚩	Environmental Sounds Automatic Classification	Marcelo Freitas	marcelo@gmail.com	forest.wav	result_forest.png	10/07/2023, 16:08
✓	Environmental Sounds Automatic Classification	Marcelo Freitas	marcelo@gmail.com	result_test.wav	result_result_test.png	10/07/2023, 16:00
✓	Environmental Sounds Automatic Classification	Marcelo Freitas	marcelo@gmail.com	forest.wav	result_forest.png	10/07/2023, 15:59

FMdeploy - ReactJS



Run History

Flag	Project	Name	Email	Input	Output	Time
	Decoding Sound Events Related to ASD - csv	Marcelo Freitas	marcelo@gmail.com	forest.wav	result_forest.csv	11/07/2023, 17:36
	Decoding Sound Events Related to ASD - csv	Marcelo Freitas	marcelo@gmail.com	forest.wav	result_forest.csv	11/07/2023, 13:25

Output not flagged

Flag erroneous output

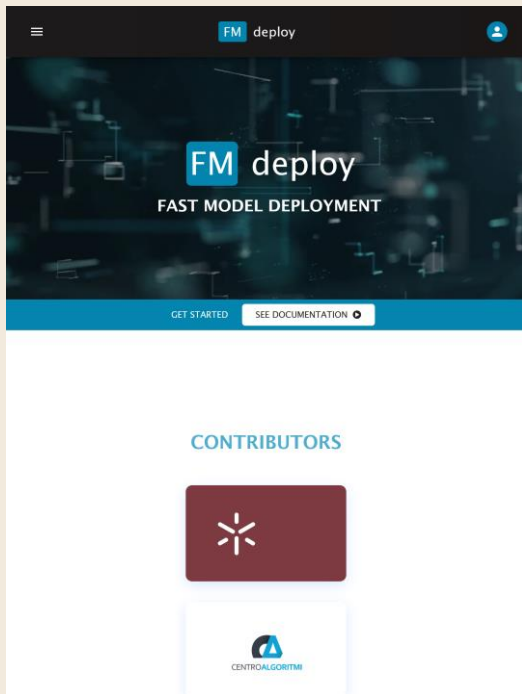
Flag an output if you notice something wrong or unexpected, and describe the issue or error you have observed in the provided text box.

Description

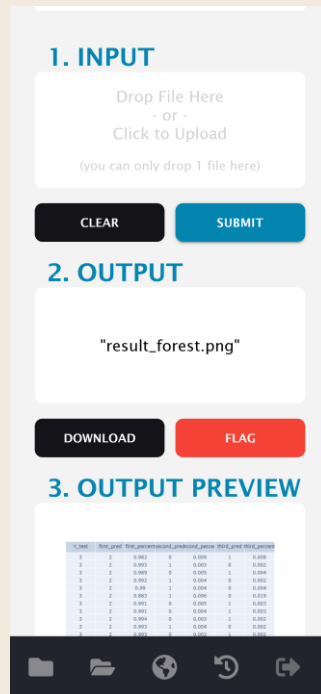
SUBMIT

	Decoding Sound Events Related to ASD - csv	Marcelo Freitas	marcelo@gmail.com	forest.wav	result_forest.csv	11/07/2023, 13:18
---	--	-----------------	-------------------	----------------------------	-----------------------------------	-------------------

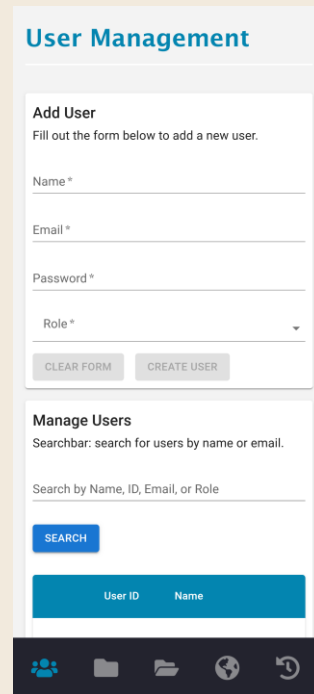
Responsive Design



UI Ipad



UI Iphone 11 Pro Max



UI Samsung Galaxy S20



03

ML Models

Deployed ML Models



ML Models Deployed



ML Model 02

- Deep Learning for Real-Time Decoding of Sound Events Related to Autism
- Diana Martins Dissertation
- University of Minho, 2021

ML Model 01

- Automatic Segmentation of the Olfactory Bulb
- Francisca Assunção Dissertation
- University of Minho, 2021

ML Model 03

- Environmental Sounds Automatic Classification
- Rafaela Machado Dissertation
- University of Minho, 2021



04

Conclusions

Findings and Research Opportunities



Conclusions

FMdeploy simplifies ML model deployment via a FastAPI-driven API and a React-powered web app.

Enhanced accessibility, direct model feedback, secure HTTPS communication, and reproducibility are key features.

Successful deployment of three models: Automatic Segmentation of the Olfactory Bulb, Decoding of Sound Events related to ASD, and Environmental Sounds Automatic Classification.



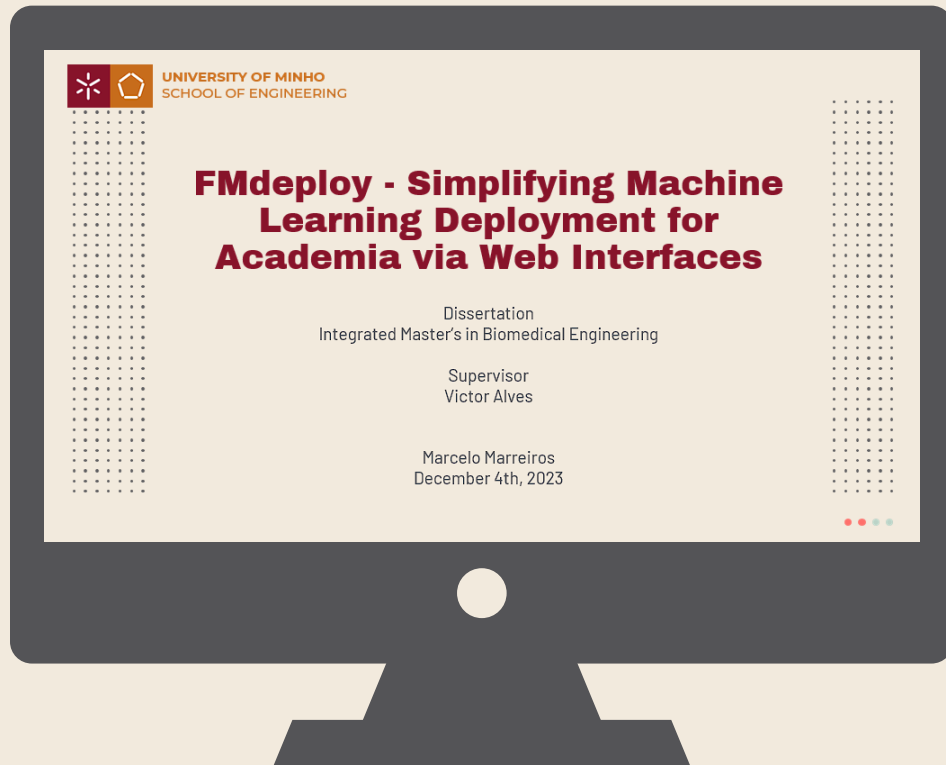
Research Opportunities



Explore the deployment of diverse ML models using varied libraries, inputs, and outputs to broaden platform support for different applications and domains.

Perform thorough evaluations of the developed API and web application for performance and effectiveness, involving targeted users for valuable feedback. Stress tests can assess system robustness and scalability under heavy workloads. User surveys and questionnaires can gather insights for iterative improvements.

Consider developing a mobile application using React Native, leveraging the existing React-based web interfaces to efficiently build for both Android and iOS platforms.



THANK YOU FOR LISTENING!

Supervisor

Victor Alves

Student

Marcelo Marreiros